

Advanced Computer Networks – PCS223

ASSIGNMENT 1

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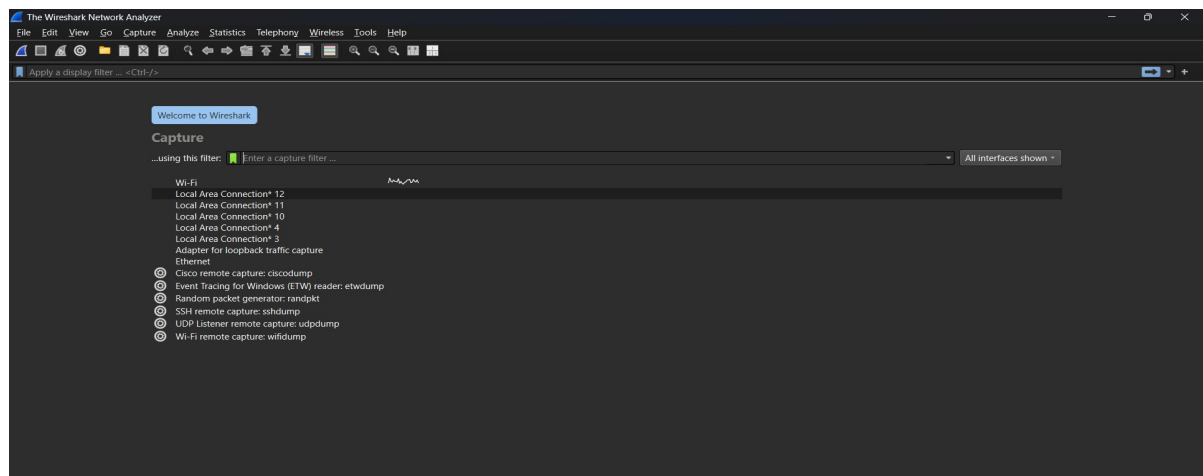
Subgroup – CS2

Roll No. – 8025320034

1. To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

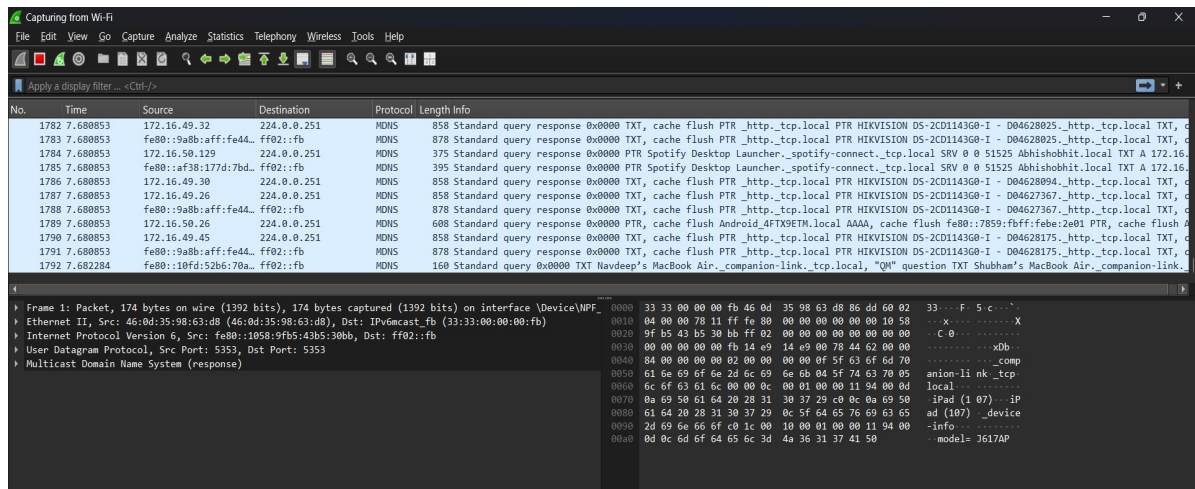
Step 1: Launch Wireshark

- Open **Wireshark**
- You will see a list of **network interfaces**
 - Wi-Fi
 - Ethernet



Step 2: Start Packet Capture

- Double-click on **Wi-Fi** (or active interface)
- Packet capture will start immediately



Step 3: Generate Network Traffic

Perform any of the following:

- Open a web browser
- Visit any website (e.g., google.com)
- Open Command Prompt and run:
- ping google.com

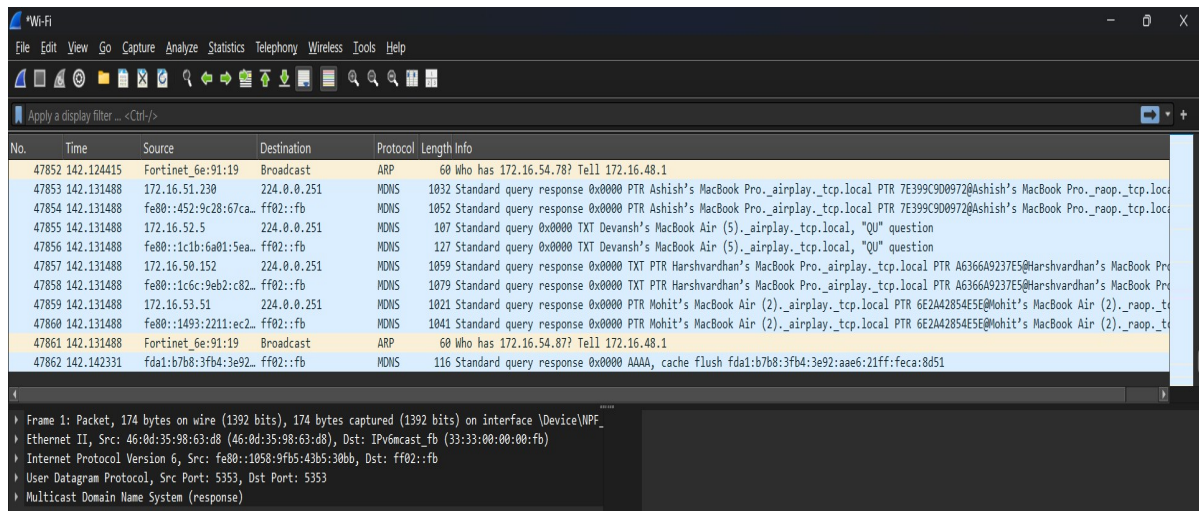
```
C:\Users\dell9>ping google.com

Pinging google.com [142.250.182.238] with 32 bytes of data:
Reply from 142.250.182.238: bytes=32 time=53ms TTL=118
Reply from 142.250.182.238: bytes=32 time=14ms TTL=118
Reply from 142.250.182.238: bytes=32 time=18ms TTL=118
Reply from 142.250.182.238: bytes=32 time=32ms TTL=118

Ping statistics for 142.250.182.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 53ms, Average = 29ms
```

Step 4: Stop Capture

- Click the **red Stop button**
- Captured packets will remain on the screen



Step 5: Apply Basic Filters

Use the **Display Filter** bar:

Filter

Purpose

ip

Show IP packets

Show TCP

tcp

packets

Show UDP

udp

packets

Show DNS

dns

packets

Show ping

icmp

packets

icmp						
No.	icmp	Source	Destination	Protocol	Length	Info
→ 3111 6.380559	icmpv6	172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 3127)
← 3127 6.380559		172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=118 (request in 3111)
3513 7.342610		172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 3518)
3518 7.364507		172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=118 (request in 3513)
4290 8.357826		172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 4323)
4323 8.394766		172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=118 (request in 4290)
4906 9.374616		172.16.41.135	172.217.24.238	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 4911)
4911 9.402421		172.217.24.238	172.16.41.135	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=118 (request in 4906)

Step 6: Analyze a Packet

- Click on any packet
- Observe:
 - **Ethernet header**
 - **IP header**
 - **TCP/UDP header**
 - Payload data

Wireshark - Packet 4906 - Wi-Fi	
▶ Frame 4906: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{30E8F03D-A225-4923-BF48-48BA90E0F267}, id 0 ▼ Ethernet II, Src: Intel_0a:fe:60 (98:43:fa:0a:fe:60), Dst: Fortinet_6e:91:0f (04:d5:90:6e:91:0f) ▶ Destination: Fortinet_6e:91:0f (04:d5:90:6e:91:0f) ▶ Source: Intel_0a:fe:60 (98:43:fa:0a:fe:60) - Type: IPv4 (0x0800) [Stream index: 4] ▼ Internet Protocol Version 4, Src: 172.16.41.135, Dst: 172.217.24.238 0100 = Version: 4 0101 = Header Length: 20 bytes (5) ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) - Total Length: 60 - Identification: 0x2c90 (11408) 0000 = Flags: 0x0 ...0 0000 0000 0000 = Fragment Offset: 0 - Time to Live: 128 - Protocol: ICMP (1) - Header Checksum: 0x0000 [validation disabled] [Header checksum status: Unverified] - Source Address: 172.16.41.135 - Destination Address: 172.217.24.238 [Stream index: 177] ▶ Internet Control Message Protocol	<pre> 0000 04 d5 90 6e 91 0f 98 43 fa 0a fe 60 08 00 45 00 ...n..C..E 0010 00 3c 2c 90 00 00 80 01 00 00 ac 10 29 87 ac d9 <.....).... 0020 18 ee 08 00 4d 57 00 01 00 04 61 62 63 64 65 66 ...MW...abcdef 0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv 0040 77 61 62 63 64 65 66 67 68 69 wabcdefg hi </pre>



Observation

Interface used:

- Protocols observed:

- Source and destination IP addresses
 - Packet length
-

Result

Live network packets were successfully captured using Wireshark, and basic network protocols were identified and analyzed.